LOVELY PROFESSIONAL UNIVERSITY

Academic Task-2

School of Computer Science and Engineering	Faculty of Technology and Sciences
Course Code: CSE 316	Course Title: Operating Systems
Term: 25261	Max. Marks:30

Instructions for Assignment Submission

1. This assignment is a compulsory CA component.

2. The assignment (project) is a group activity, but submission on UMS will be individual for each student.

3. The project submission mode is **Online** only. The student has to upload the report on or before the last date on UMS only. No submission via e-mail or pen-drive, or any media will be accepted.

4. Non-submission of project report on **UMS** till the last date will result in **ZERO** marks.

5. The student is supposed to code the project on his/her own. If it is discovered at any stage that the student has used unfair means, such as copying from peers or copying and pasting code directly taken from the internet, etc. **ZERO** marks will be awarded to the student.

6. The student who shares his assignment with other students (either in the same section or a different section) will also get **ZERO** marks.

Phase 1: Problem Assignment

Guidelines For Students:

Step 1: AI-Guided Development

AI Guidance:

- Students will submit the problem statement to the AI system.
- AI will analyze the problem and provide a **detailed explanation** of the project statement along with its breakdown.

The prompt that will be asked by a student to AI:

I have been assigned the following problem statement:

"Paste your Problem Statement Here."

Provide the following details to help me implement this project:

1. *Project Overview: Explain the project's goals, expected outcomes, and scope.*
2. *Module-Wise Breakdown: Divide the project into 3 separate modules (e.g., GUI, ML, data visualization), explaining their purposes and roles.*
3. *Functionalities: List key features for each module, with examples for clarity.*
4. *Technology Recommendations: Suggest programming languages, libraries, and tools to use.*
5. *Execution Plan: Provide a step-by-step implementation guide with tips for efficiency.*

Implementation Language: Any as per your project requirement

Step 2: Revision Tracking on GitHub

GitHub Workflow:

- Students will:
    - Create a repository for the project.
    - Maintain at least 7 revisions and push changes regularly with clear commit messages.
    - Use branches for feature implementation and merge them into the main branch after testing.

Step 3: Report Format

1. Project Overview

2. Module-Wise Breakdown

3. Functionalities

4. Technology Used

- Programming Languages:
- Libraries and Tools:
- Other Tools: [e.g., GitHub for version control]

5. Flow Diagram

6. Revision Tracking on GitHub

- Repository Name: [Insert Repository Name]
- GitHub Link: [Insert Link]

7. Conclusion and Future Scope

8. References

Appendix

A. AI-Generated Project Elaboration/Breakdown Report

[Paste the AI-generated breakdown of the project in detail]

B. Problem Statement: [Paste Problem Statement]

C. Solution/Code:

[Paste complete code of entire project here]

Phase 2: Project Evaluation (30 marks)

The entire evaluation has been divided into 2 steps:

Step 1: AI Evaluation (15 marks)

The faculty members will use AI to evaluate the project submitted by the student. This AI evaluation will be out of 15 marks.

Step 2: Presentation and Viva (15 marks)

The faculty members will take the presentation and viva of the students in their respective classes and will evaluate the students as per the shared rubrics.

Rubric for Presentation and Viva

Criterion	**Details**	**Marks**
Understanding of Project and Question response	Depth of knowledge about the problem statement, implementation, and challenges.	**10**
- Excellent	Demonstrates thorough understanding with clear and concise answers.	7
- Good	Good understanding but some minor gaps.	5
- Needs Improvement	Limited understanding and vague answers.	3
Problem-Solving Approach	Innovative solutions and effective handling of challenges.	**3**
- Excellent	Showcases innovation and robust problem-solving skills.	3

Criterion	**Details**	**Marks**
- Good	Reasonable problem-solving with minor shortcomings.	2
- Needs Improvement	Basic or ineffective problem-solving approach.	1
Communication Skills	Clarity and confidence in explaining the project.	2
- Excellent	Articulates ideas effectively with high confidence.	**2**
- Good	Reasonably clear explanations with some hesitation.	1
- Needs Improvement	Difficulty in articulating thoughts or lack of confidence.	0

Project Title: Intelligent CPU Scheduler Simulator

Leader: Pranjal Kumar (12501652)
Group Members: Akanchha Bhardwaj , Ansh Kumar Pandit (12526245 , 12503147)
Section: K24EL
Description: Develop a simulator for CPU scheduling algorithms (FCFS, SJF, Round Robin, Priority Scheduling) with real-time visualizations. The simulator should allow users to input processes with arrival times, burst times, and priorities and visualize Gantt charts and performance metrics like average waiting time and turnaround time.

Project Title: Dynamic Memory Management Visualizer

Leader: Vamsi ram salla (12507380)
Group Members: Vamsi ram salla , Rohit salapu , Jyothi prakash malla (12507390 ,12524731)
Section: K24EL
Description: Build a tool to simulate and visualize memory management techniques like paging, segmentation, and virtual memory. The system should handle user-defined inputs for memory allocation, page faults, and replacement algorithms (FIFO, LRU).

Project Title: Inter-Process Communication (IPC) Debugger

Leader: Deepansh Tank (12503753)
Group Members: Arpit Khandelwal (12503747)
Section: K24EL
Description: Design a debugging tool for inter-process communication methods (pipes, message queues, shared memory) to help developers identify issues in synchronization and data sharing between processes. Include a GUI to simulate data transfer and highlight potential bottlenecks or deadlocks.

Project Title: Deadlock Prevention and Recovery Toolkit

Leader: Sahil (12503888)
Group Members: Aryan Kumar Thakur, Ashutosh kumar singh (12517018, 12523734)
Section: K24EL
Description: Create a toolkit that detects, prevents, and recovers from deadlocks in real-time. The system should implement algorithms like Banker's Algorithm and display resource allocation graphs for clarity. Include a feature for simulating deadlock scenarios with custom user inputs.

Project Title: Secure File Management System

Leader: Harsh Sokhal (12524316)
Group Members: Vaishali, Narendra Yogesh Ahire (12503189, 12524755)
Section: K24EL
Description: Develop a secure file management system that incorporates authentication mechanisms (password-based, two-factor), protection measures (access control, encryption), and detection of common security threats (buffer overflow, malware). Users should be able to perform file operations like read, write, share, and view metadata securely.

Project Title: Real-Time Multi-threaded Application Simulator

Leader: Dipanjan Das (12501265)
Group Members: Krishnakant Yadav, Ankush Kumar. (12523006, 12527152)
Section: K24EL

Description: Develop a simulator to demonstrate multithreading models (Many-to-One, One-to-Many, Many-to-Many) and thread synchronization using semaphores and monitors. The simulator should visualize thread states and interactions, providing insights into thread management and CPU scheduling in multi-threaded environments.

Project Title: Virtual Memory Optimization Challenge

Leader: Souvik Mondal (12500610)
Group Members: Ali Asgar, Bipul Mondal (12505331, 12502488)
Section: K24EL
Description: Create a virtual memory management tool that visualizes the behavior of paging and segmentation, including page faults and demand paging. Allow users to experiment with custom inputs for memory allocation, simulate memory fragmentation, and evaluate page replacement algorithms like LRU and Optimal.

Project Title: File System Recovery and Optimization Tool

Leader: Vinay Raut (12503569)
Group Members: Ajinkya Pawar (12521499)
Section: K24EL
Description: Design a tool for recovering and optimizing file systems. Implement methods for free-space management, directory structures, and file access mechanisms. Simulate real-world scenarios like disk crashes and provide recovery techniques while optimizing file read/write times.

Project Title: Security Vulnerability Detection Framework

Leader: Deepak Kumar Yadav (12519231)
Group Members: nan (nan)
Section: K24EL
Description: Develop a framework to detect and mitigate security vulnerabilities in operating systems, such as buffer overflows, trapdoors, and cache poisoning. The framework should simulate attacks, provide real-time detection alerts, and suggest recovery or prevention measures.

Project Title: Advanced Disk Scheduling Simulator

Leader: Aquib Jawaid Ansari (12508688)
Group Members: Aquib Jawaid Ansari, Prathamesh Gadewar, Prabhjot Singh (12508688, 12500420, 12508327)
Section: K24EL
Description: Build a disk scheduling simulator to visualize algorithms like FCFS, SSTF, SCAN, and C-SCAN. Users should be able to input custom disk access requests and see how the algorithms minimize seek time. Include performance metrics like average seek time and system throughput.

Project Title: Adaptive Resource Allocation in Multiprogramming Systems

Leader: Amiya Roy (12503247)
Group Members: Nayan Sachdeva,Mohd azaan javed (1252142612510970)
Section: K24EL
Description: Develop a system that dynamically adjusts resource allocation among multiple programs to optimize CPU and memory utilization. The solution should monitor system performance and reallocate resources in real-time to prevent bottlenecks.

Project Title: User-Friendly System Call Interface for Enhanced Security

Leader: Rishav Raj (12525964)
Group Members: Rishav Raj, Piyush, Aditya Yadav (12525964, 12521121, 12503771)
Section: K24EL
Description: Design an intuitive interface for system calls that enhances security by preventing unauthorized access. The interface should include authentication mechanisms and provide detailed logs of system call usage.

Project Title: Real-Time Process Monitoring Dashboard

Leader: Santosh Kumar (12525269)
Group Members: Kanhaiya Kumar, Satwik (12500673, 12529164)
Section: K24EL
Description: Create a graphical dashboard that displays real-time information about process states, CPU usage, and memory consumption. The tool should allow administrators to manage processes efficiently and identify potential issues promptly.

Project Title: Energy-Efficient CPU Scheduling Algorithm

Leader: Shubham Yadav (12500406)
Group Members: Naureen, Rashmi (1250257712519170)
Section: K24EL
Description: Develop a CPU scheduling algorithm that minimizes energy consumption without compromising performance. The algorithm should be suitable for mobile and embedded systems where power efficiency is critical.

Project Title: Scalable Thread Management Library

Leader: Ashutosh Kumar (12507944)
Group Members: Arjun Baliyan, Manish Kumar Singh (12502688, 12502989)
Section: K24EL
Description: Design a thread management library that supports efficient creation, synchronization, and termination of threads. The library should be scalable to handle thousands of threads concurrently, suitable for high-performance computing applications.

Project Title: Automated Deadlock Detection Tool

Leader: Soni Kumari (12520804)
Group Members: Soni Kumari, Abhijeet Kumar (12520804, 12524986)
Section: K24EL
Description: Develop a tool that automatically detects potential deadlocks in system processes. The tool should analyze process dependencies and resource allocation to identify circular wait conditions and suggest resolution strategies.

Project Title: Secure Authentication Module for Operating Systems

Leader: Sahil kumar (12503343)
Group Members: Sahil kumar, kotha purna rathna mahiram, Soda Charan teja (nan)
Section: K24EL
Description: Create a robust authentication module that integrates with existing operating systems to enhance security. The module should support multi-factor authentication and protect against common vulnerabilities like buffer overflows and trapdoors.

Project Title: Efficient Page Replacement Algorithm Simulator

Leader: Raj Aryan (12508555)
Group Members: Rahul kumar , Md Safdar (12526496, 12529162)
Section: K24EL
Description: Design a simulator that allows users to test and compare different page replacement algorithms (FIFO, LRU, Optimal). The simulator should provide visualizations and performance metrics to aid in understanding algorithm efficiency.

Project Title: Distributed File System with Fault Tolerance

Leader: Mohd Anas (12517504)
Group Members: No other group member (No other group member)
Section: K24EL
Description: Develop a distributed file system that ensures data availability and integrity across multiple nodes. The system should incorporate fault tolerance mechanisms to handle node failures gracefully.